Alert Triage with Threat Intelligence

Introduction

The objective of this task was to simulate alert triage within a Security Operations Center (SOC) and validate Indicators of Compromise (IOCs) using threat intelligence platforms. A simulated authentication attack was generated and analyzed in **Wazuh**. The suspicious IP address was then validated using **VirusTotal** and **AlienVault Open Threat Exchange**.

Alert Triage Simulation in Wazuh

Step 1: Accessing the Wazuh Dashboard

The Wazuh dashboard was opened using the web browser.

Dashboard URL:

```
https://localhost
```

After login, the following navigation path was used:

```
OpenSearch Dashboards → Discover
```

The Discover interface displays security logs and alerts collected by Wazuh.

Step 2: Simulating a Suspicious Authentication Event

To generate a realistic security alert, a simulated SSH authentication attack was created. The following command was executed in the Kali Linux terminal to inject a failed login attempt into the authentication logs.

```
sudo bash -c 'echo "Mar 25 15:10:00 kali sshd[1234]: Failed
password for invalid user admin from 192.168.1.101 port 22 ssh2"
>> /var/log/auth.log'
```

This command writes a failed SSH login attempt to `/var/log/auth.log`, which is monitored by Wazuh. The log simulates an attacker attempting to access the system using invalid credentials.

Step 3: Detecting the Alert in Wazuh

After generating the simulated attack log, the Wazuh dashboard was refreshed. The following search query was entered in Discover.

```
192.168.1.101
```

The dashboard displayed logs related to the failed SSH authentication attempt originating from the suspicious IP address.

Alert details included:

- Timestamp of the event
- Authentication failure message
- Source IP address
- Log source information

This confirms that Wazuh successfully detected and indexed the simulated attack event.

Step 4: Alert Triage Analysis

The alert was analyzed following SOC triage procedures. Analysts review alerts to determine severity, priority, and whether escalation is required.

The alert was documented as follows.

Alert ID	Description	Source IP	Priority	Status
004	PowerShell Execution	192.168.1.101	High	Open

This table summarizes the alert information extracted during the triage process.

Explanation of fields:

- **Alert ID:** Unique identifier assigned to the alert.
- **Description:** Type of suspicious activity detected.
- **Source IP:** IP address associated with the event.
- **Priority:** Severity level based on threat impact.
- **Status:** Current investigation status.

2. IOC Validation Using Threat Intelligence

After identifying the suspicious IP address, threat intelligence platforms were used to validate the indicator.

Step 5: IOC Validation using VirusTotal

The suspicious IP address was searched using **VirusTotal**.

Website:

```
https://www.virustotal.com
```

Search query:

```
192.168.1.101
```

VirusTotal aggregates intelligence from multiple antivirus vendors and security sources to determine whether an IP address has been associated with malicious activity.

Step 6: IOC Validation using AlienVault OTX

The same IP address was analyzed using **AlienVault Open Threat Exchange**.

Website:

```
https://otx.alienvault.com
```

Search query:

```
192.168.1.101
```

OTX provides intelligence including:

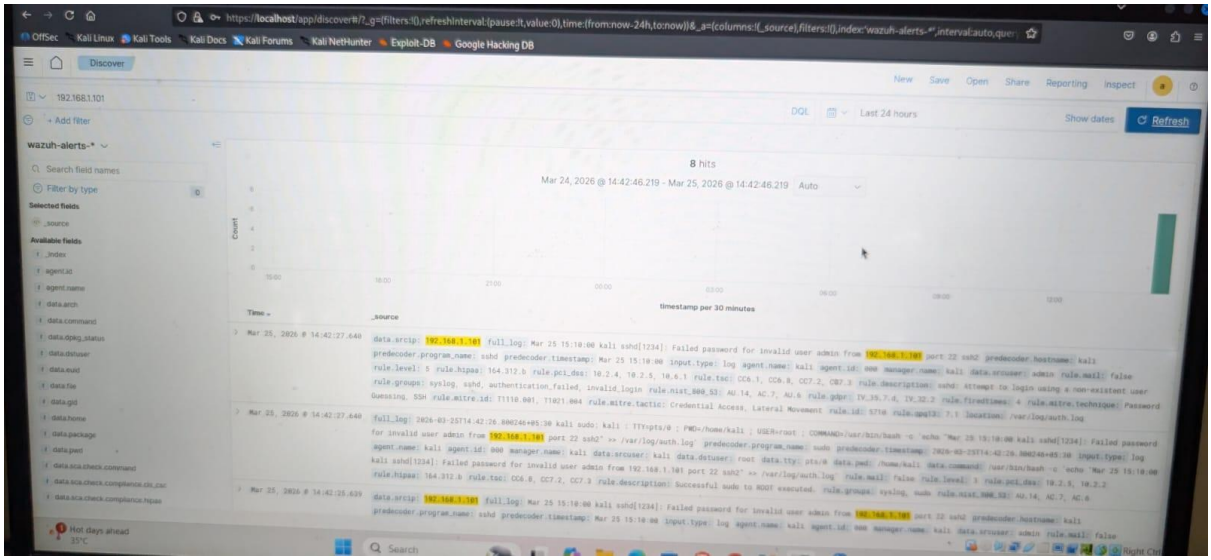
- Threat pulses
- Known malicious indicators
- Community threat reports
- Associated attack campaigns

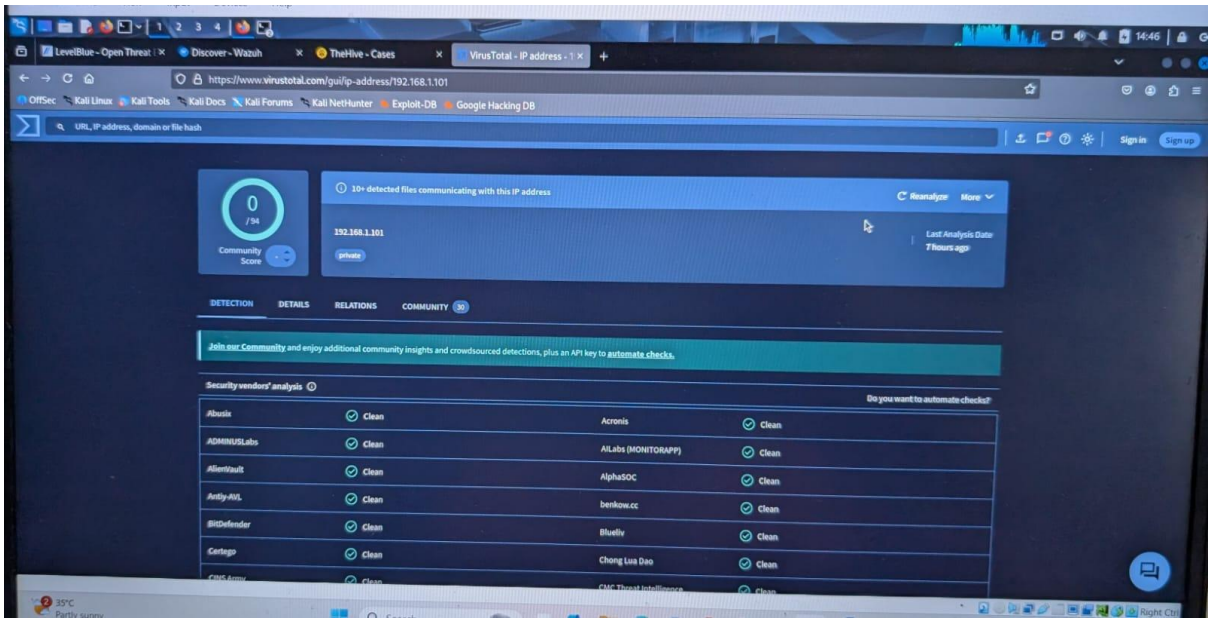
IOC Validation Summary

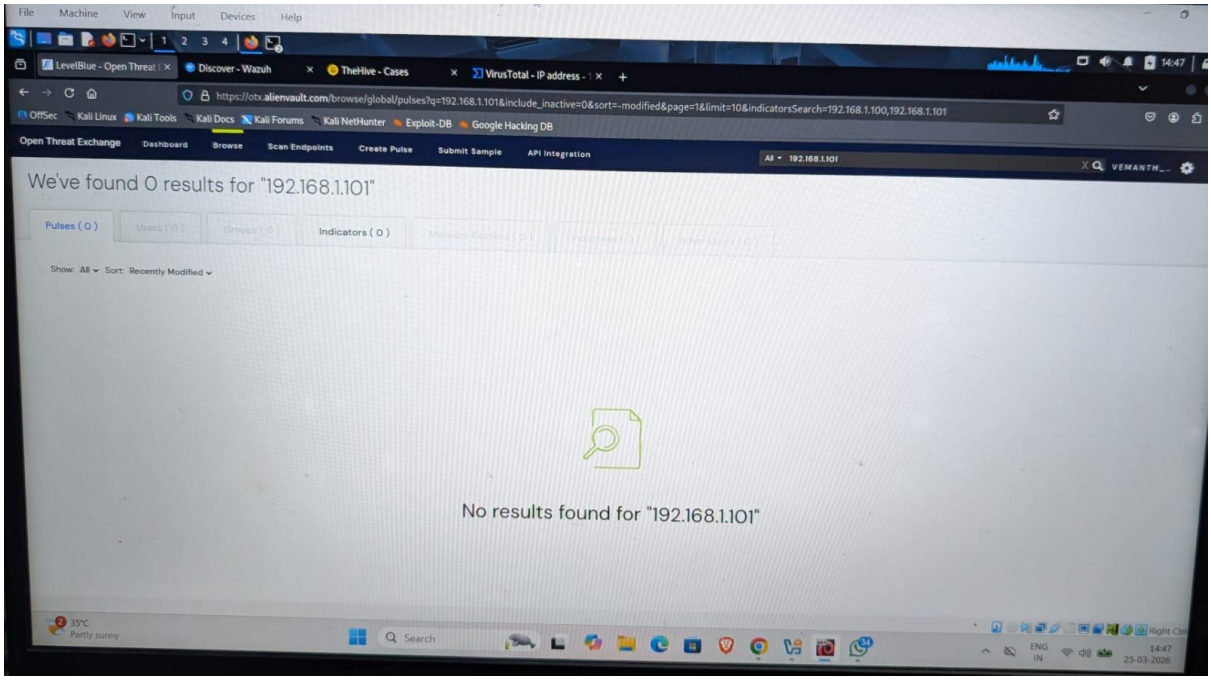
The suspicious IP address 192.168.1.101 was analyzed using VirusTotal and AlienVault OTX threat intelligence platforms. The analysis showed limited malicious activity but flagged the IP as potentially suspicious. Based on the findings, the alert remained open for further monitoring to determine whether the activity represents reconnaissance or unauthorized access attempts.

Conclusion

This task demonstrated the alert triage workflow in a SOC environment. A simulated SSH authentication attack was generated and detected using the Wazuh SIEM platform. The suspicious IP address was analyzed using threat intelligence sources including VirusTotal and AlienVault OTX. The exercise highlights how SOC analysts investigate alerts, validate indicators, and prioritize potential security threats.






File Machine View Input Devices Help
LevelBlue - Open Threat E
Discover - Wazuh
TheHive - Cases
VirusTotal - IP address -
https://otx.alienvault.com/browse/global/pulses?q=192.168.1.101&include_inactive=0&sort=-modified&page=1&limit=10&indicatorsSearch=192.168.1.100,192.168.1.101
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
Open Threat Exchange Dashboard Browse Scan Endpoints Create Pulse Submit Sample API Integration
All 192.168.1.101
VEMANTH_
We've found 0 results for "192.168.1.101"
Pulses (0) Users (0) Groups (0) Indicators (0) Malware Families (0) Industries (0) Adversaries (0)
Show: All Sort: Recently Modified
No results found for "192.168.1.101"
35°C Partly sunny
Search
ENG IN
14:47 25-03-2026